

PHYSIOL 578: HW1

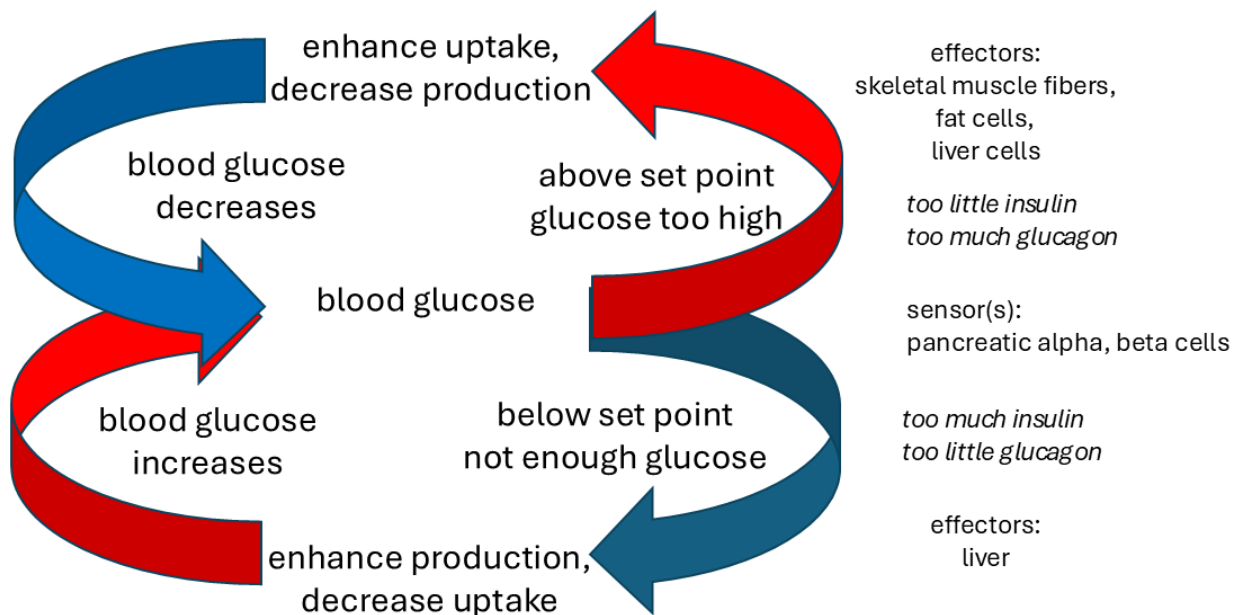
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- 1 Describe a physiologic homeostatic system. Do not use an example from class. If you work with others, your answer should be unique (each student should choose a unique homeostatic system). A drawing may be very helpful. Identify the key components of the homeostatic system and describe briefly how they function and interact. Use a single page for your response.

In blood glucose homeostasis, pancreatic alpha and beta cells monitor glucose levels in the bloodstream. If glucose levels are too high, pancreatic beta cells release the hormone insulin, which signals skeletal muscle fibers and fat cells to take up glucose and promotes glucose storage in liver cells. These responses lower the concentration of glucose in the bloodstream.

Conversely, if blood glucose levels fall too low, beta cells reduce insulin release, and alpha cells are stimulated to release glucagon. Glucagon raises blood glucose by signaling the liver to perform glycogenolysis and gluconeogenesis.



2 Check the hummingbird folder in this section. Explain how a hummingbird can fly across the Gulf of Mexico.

3 Ohm's Law: explain how ion channels affect input resistance.

In a cell, ionic current is given by

$$I = g(V - E_{\text{rev}})$$

where:

I is the ionic current

g is conductance (inverse of resistance $1/R$)

$V - E_{\text{rev}}$ is the driving potential

The driving potential $V - E_{\text{rev}}$ is generated by the electrochemical gradient across the membrane. Because the membrane is relatively impermeable to ions, the resistance R is very high (conductance g is low) and there is hardly any ions crossing the membrane, thus minimal ionic current I , even when there is a driving potential. However, with ion channels that allow ions to pass through the membrane according to the driving potential, the current I increases and **resistance R decreases**.

4 The Nernst Equation: calculate the reversal potential for a channel permeable to a single ion. Name the ion and the concentrations you chose and why you chose them ("because I found them in a book" is not a good answer). Be sure to list all parameters.

Nernst Equation is given by

$$E_{\text{rev}} = \frac{RT}{zF} \ln \frac{[\text{Ion}]_{\text{out}}}{[\text{Ion}]_{\text{in}}}$$

where:

R is the universal ideal gas constant

T is temperature in kelvins

z is the charge

F is Faraday's constant

$[\text{Ion}]$ is the concentration of the ion

Kir2.1 is an inward-rectifier potassium channel that is permeable mostly to K^+ . Thus its reversal potential E_{rev} can be calculated by:

$$\begin{aligned} E_{\text{rev}} &= \frac{RT}{(1)F} \ln \frac{4 \times 10^{-3}\text{M}}{1.35 \times 10^{-1}\text{M}} \\ &= \frac{(8.314\text{J}/(\text{mol} \cdot \text{K}))(310.15\text{K})}{(1)(9.648 \times 10^4\text{C}/\text{mol})} (-3.519) = \frac{2.578 \times 10^3\text{J}/\text{mol}}{9.648 \times 10^4\text{C}/\text{mol}} (-3.519) \\ &= -0.094\text{J}/\text{C} = \boxed{-94\text{mV}} \end{aligned}$$

where:

$T = 310.15\text{K}$ as the temperature of the body

$z = +1$ as K^+ has charge of $+1$

$[\text{Ion}]_{\text{in}} = 1.35 \times 10^{-1}\text{M}$ as the normal intracellular concentration of K^+ for a mammalian neuron

$[\text{Ion}]_{\text{out}} = 4 \times 10^{-3}\text{M}$ as the normal extracellular concentration of K^+ for a mammalian neuron

Therefore, for a Kir2.1 channel in a mammalian neuron with 135mM intracellular K^+ and 4mM extracellular K^+ at 37°C , the predicted reversal potential is $\boxed{-94\text{mV}}$.

5 GHK equation: use physiologic concentrations appropriate for the squid giant axon. Estimate the resting potential of the axon. What might complicate your answer? Be sure to list all parameters you used in your calculation.